torch.jit.annotate#

https://pytorch.org/docs/stable/generated/torch.jit.annotate.html

Description:

Use to give type of the_value in TorchScript compiler. This method is a pass-through function that returns the_value, used to hint TorchScript compiler the type of the_value. It is a no-op when running outside of TorchScript. Though TorchScript can infer correct type for most Python expressions, there are some cases where type inference can be wrong, including: Empty containers like [] and {}, which TorchScript assumes to be container of Tensor Optional types like Optional[T] but assigned a valid value of type T, TorchScript would assume it is type T rather than Optional[T]

Function Signature

```
torch.jit.annotate(the_type, the_value)[source]#
Parameters
Returns
```

Returns:

the_value is passed back as return value.

Code Examples

```
import torch
from typing import Dict

@torch.jit.script
def fn():
    # Telling TorchScript that this empty dictionary is a (str -
> int) dictionary
    # instead of default dictionary type of (str -> Tensor).
    d = torch.jit.annotate(Dict[str, int], {})

    # Without `torch.jit.annotate` above, following statement
would fail because of
    # type mismatch.
    d["name"] = 20
```

Detailed Documentation

Documentation

Use to give type of the_value in TorchScript compiler.

This method is a pass-through function that returns the_value, used to hint TorchScript compiler the type of the_value. It is a no-op when running outside of TorchScript.

Though TorchScript can infer correct type for most Python expressions, there are some cases where type inference can be wrong, including:

Empty containers like [] and {}, which TorchScript assumes to be container of Tensor

Optional types like Optional[T] but assigned a valid value of type T, TorchScript would assume it is type T rather than Optional[T]

Note that annotate() does not help in __init__ method of torch.nn.Module subclasses because it is executed in eager mode. To annotate types of torch.nn.Module attributes, use Attribute() instead.

the_type – Python type that should be passed to TorchScript compiler as type hint for the_value

the_value – Value or expression to hint type for.

the_value is passed back as return value.